Homework #2
Due Friday January 24
Submit to Canvas

Read Chapters 3 and 5 of Strogatz

The following problems are from Strogatz Second Edition, the one with the black cover. Scans of the exercises in chapters 3 and 5 can be found on Canvas in the files section, in a folder called homeworks.

Solutions to the following 8 problems should be written up carefully and turned in online. Note that problems with a * are worth 10 points; all unstarred problems are worth 5 points. (Full assignment is worth 50 points.)

Strogatz: 3.4.9, 3.4.15, 3.6.2*, 3.7.6* (a)-(j) (Note: use $z(0)=0$)

Silber: Consider the system of linear differential equations

$$dx/dt = -x + y$$
$$dy/dt = r x - y,$$

where r is a real constant. Find a value of r for which the origin is a (i) stable spiral, (ii) stable node, (iii) saddle point, (iv) on the boundary between stable spiral and stable node, and (v) on the boundary between stable node and saddle. For each of the chosen values of r plot a phase portrait in the (x,y)-phase plane. (An option is to do the latter using Wolfram Alpha/Mathematica's StreamPlot function.)

Strogatz: 5.1.10, 5.2.2, 5.2.11

Extra Credit Problem (worth 5 points): complete 3.7.6. (Note that I only assigned part of that problem above; I omitted 3.7.6 (k) in the required part.)

Extra Credit Problem (worth 10 points): 5.2.14*